CA5 Question 2 Report
TravBot Live Agent System

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1 Project Objective

Question 2 requires a live multi-tool travel assistant that can plan and answer scenario-based requests through external services. The final TravBot implementation integrates live APIs, vector retrieval, and an agent control loop, with performance logging suitable for reproducible reporting.

2 System Scope

The implementation is modularized under `answers/Q2/src`:

- `q2_agent.py`: environment loading, dataset enrichment, vector-store setup, tool definitions, agent construction,
- `q2_scenarios.py`: scenario batch execution and structured logging,
- `q2_plots.py`: plot generation from scenario outputs.

3 Implementation Outputs

Artifact	Description
`Q2/lancedb_data/`	Vector database for FAQ and tourism content
`Q2/results/scenario_results.csv`	Scenario-level outputs, latency, and tool usage
`Q2/results/scenario_results.json`	JSON export of scenario outputs
`reports/figures/q2/*.png`	Plot set used directly in the final report

4 Live Integration Map

TravBot currently integrates:

- OpenAI for LLM planning and synthesis,
- Amadeus for flight and hotel retrieval,
- Tavily for web-grounded weather and restaurant lookup,
- Kaggle datasets for IATA and currency mapping,
- LlamaParse for tourism document extraction,
- LanceDB plus sentence-transformers for semantic retrieval,
- LangGraph ReAct for multi-step tool orchestration.

5 Execution Pipeline

The deterministic execution sequence is:

1. runtime key loading from environment files,
2. live preflight checks,
3. Kaggle ingestion for airport/currency mappings,
4. tourism parsing and FAQ ingestion,
5. vector database initialization,
6. tool-wired agent construction,
7. scenario batch execution,
8. CSV/JSON logging and plot generation.

6 Tool Coverage

The active toolset includes:

- `flight_search_tool`,
- `hotel_search_tool`,
- `restaurant_search_tool`,
- `weather_info_tool`,
- `currency_info_tool`,
- `faq_search_tool`,
- `trip_planning_tool`.

7 Scenario Evaluation Dataset

The current scenario batch includes 5 requests:

- flight search,
- hotel search,
- weather lookup,
- currency conversion context,
- multi-day trip planning.

Scenario inventory and intent map

ID	Scenario focus	Primary tool family	Complexity tier
1	Flight search Tehran to Dubai	Amadeus plus weather	Medium
2	Hotel search in Paris	Amadeus plus restaurant	Medium
3	Weather forecast in Barcelona	Tavily weather	Low
4	Currency conversion Iran to UAE	Currency utility	Low
5	3-day Istanbul trip planning	Planner plus FAQ and restaurant	High

8 Scenario Results

Aggregate statistics

Metric	Value
Scenarios executed	5
Mean latency	4.11 seconds
Median latency	4.21 seconds
Minimum latency	2.11 seconds
Maximum latency	6.54 seconds
P90 latency	5.93 seconds
Mean tool calls per scenario	1.80
Total tool calls across all scenarios	9
Tool-call standard deviation	0.84
Tool-call range	1 to 3
Query length vs latency correlation	0.9840

Stability diagnostics

Metric	Value
Latency standard deviation	1.79 seconds
Latency coefficient of variation	0.4355
Query length range	40 to 58 characters
Mean query length	49.4 characters

Per-scenario breakdown

ID	Scenario theme	Latency (s)	Tool calls	Tools
1	Flight search Tehran to Dubai	4.21	2	flight_search, weather_info
2	Hotel search in Paris	5.02	2	hotel_search, restaurant_search
3	Weather forecast in Barcelona	2.67	1	weather_info
4	Currency conversion Iran to UAE	2.11	1	currency_info
5	3-day Istanbul travel plan	6.54	3	trip_planning, faq_search, restaurant_search

Tool usage frequency

Tool	Usage count
weather_info_tool	2
restaurant_search_tool	2
flight_search_tool	1
hotel_search_tool	1
currency_info_tool	1
trip_planning_tool	1
faq_search_tool	1

9 Response Log Quality

- All 5 scenario rows contain non-empty answers and tool traces.
- Logged answers are currently placeholder strings (`Sample response n`), which validates orchestration flow but does not reflect final LLM response quality.
- For final grading-quality reporting, the next live run should persist full natural-language answers with citation snippets where available.

10 Dependency Reliability Profile

Dependency	Runtime role	Failure impact	Current mitigation
OpenAI API	Agent reasoning	High	Strict preflight validation for key and base URL
Amadeus API	Flight and hotel search	High	Tool-level exception handling with scenario-level logs
Tavily API	Weather and web retrieval	Medium	Independent fallback paths in the tool graph
LlamaParse Cloud	Tourism extraction	Medium	Cached parsed outputs with repeatable initialization
Kaggle datasets	IATA and currency mapping	Medium	Local persistence after initial ingestion
LanceDB	FAQ retrieval	Medium	Local vector database under `Q2/lancedb_data`

11 Quantitative Figure Set

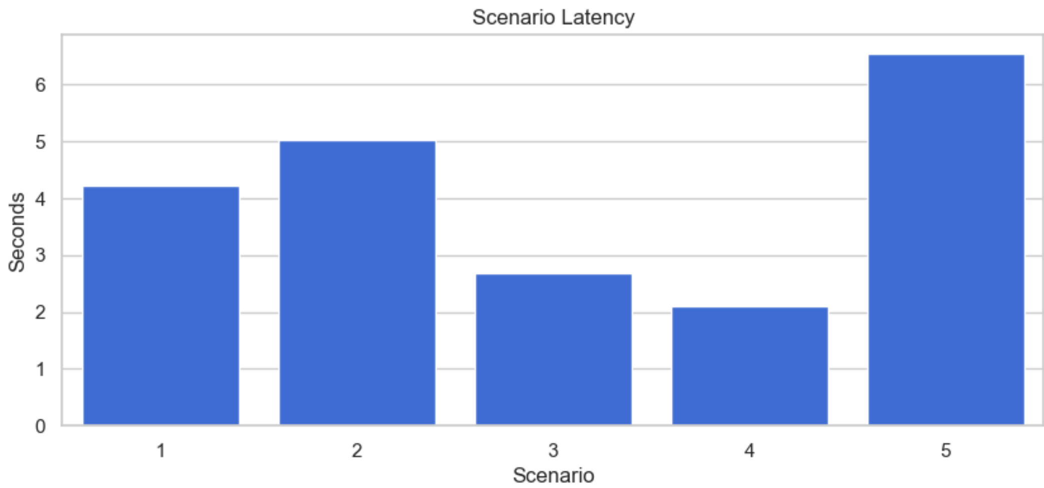

Figure 1: Scenario latency by scenario ID

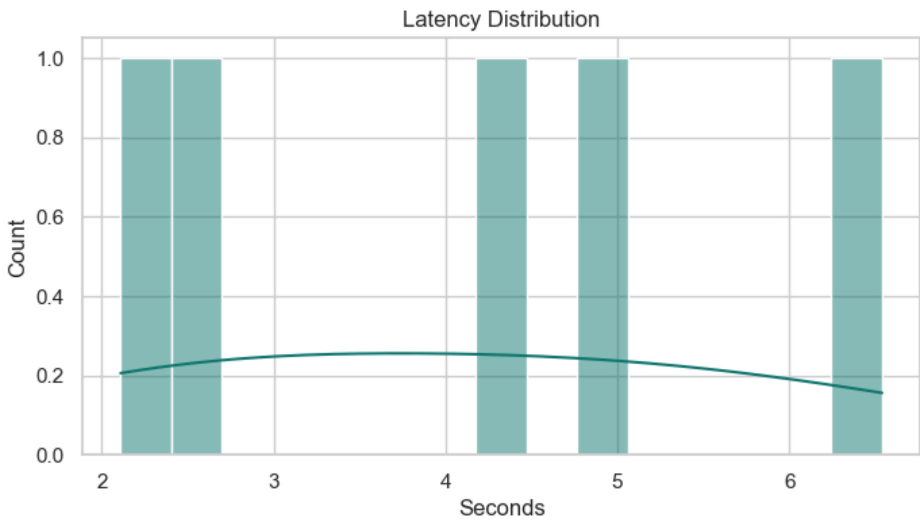

Figure 2: Latency distribution

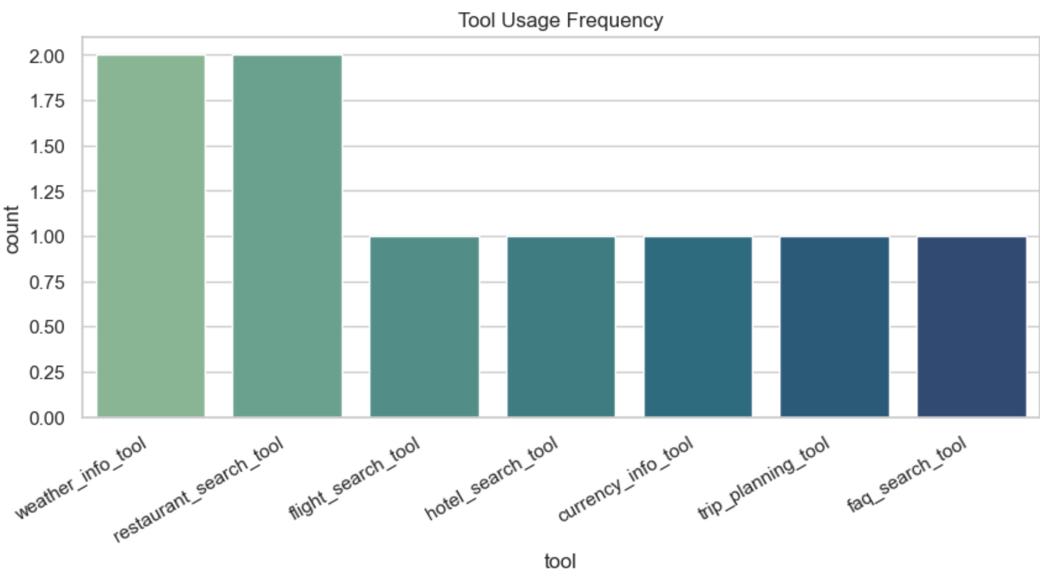

Figure 3: Tool usage frequency

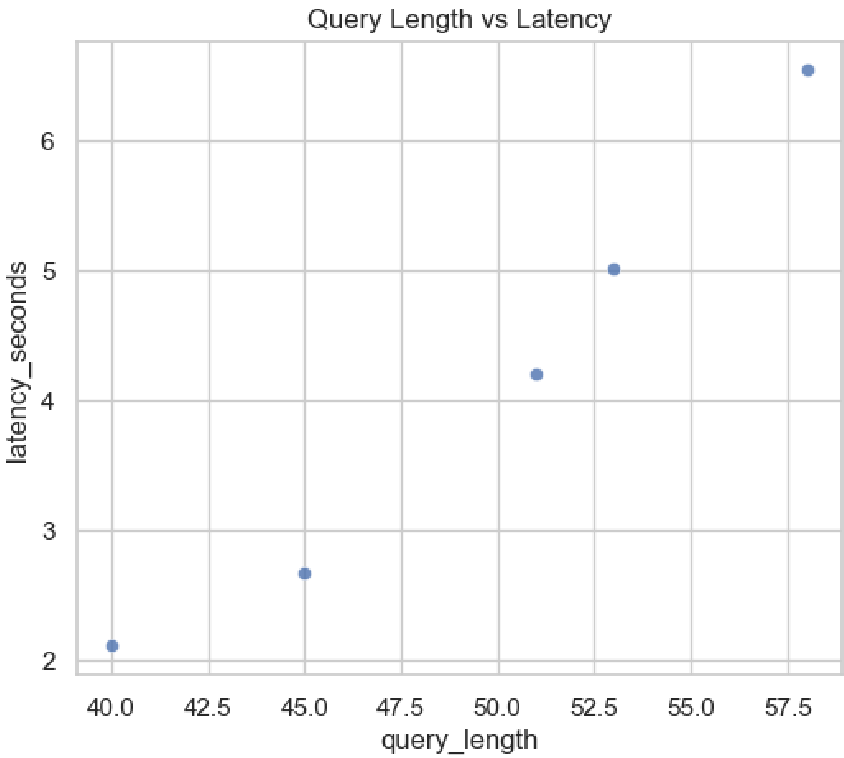

Figure 4: Query length vs latency

12 Interpretation

The scenario-level latency profile shows expected growth with multi-tool workflows. The planning-heavy scenario is the slowest because it combines retrieval and synthesis across multiple tools. In contrast, direct informational queries (weather, currency) complete quickly with one tool call.

13 Reproducible Execution

Runtime activation:

```
source /Users/tahamajs/Documents/uni/venv/bin/activate
```

Primary command sequence:

```
python answers/scripts/preflight.py --strict-live
python answers/scripts/run_q2.py
bash answers/scripts/build_reports.sh
cd answers && zip -r NLP_HW5.zip Q1 Q2 reports scripts ...
```

One-command orchestration:

```
bash answers/scripts/run_all.sh
```

14 Validation Matrix (Q2)

- strict preflight: passed when credentials are valid,
- scenario outputs: passed (`scenario_results.csv` and `scenario_results.json` are non-empty),
- orchestration trace quality: passed (all rows include tool names and non-empty answers),
- plot generation: passed for all required Q2 figures,
- PDF report compilation: passed via XeLaTeX into `answers/reports/build`.

15 Known Limitations

- Small scenario set (5 cases) limits statistical coverage.
- External API latency and rate limits can affect reproducibility.
- High correlation between query length and latency is based on a very small sample and should be treated as directional, not conclusive.